Worksheet - Chapter 9: Fuel and Adventure

Grade 5 Social Studies

A. Fill in the Blanks (30 Marks)

1. A **fuel** is any material that reacts with other substances to release heat and energy. **Answer:** fuel
2. Fuels are used for cooking, heating, running vehicles and generating **electricity**. **Answer:** electricity
3. **Wood**, coal and coke are examples of solid fuels. **Answer:** Wood
4. Petrol, diesel and **kerosene** are examples of liquid fuels. **Answer:** kerosene
5. **Natural gas** is an example of gaseous fuel. **Answer:** Natural gas
6. Fossil fuels are formed by natural processes of decomposition of buried dead **plants** and animals. **Answer:** plants
7. Coal, petroleum and **natural gas** are naturally occurring fossil fuels. **Answer:** natural gas
8. Fossil fuels take **millions** of years to form. **Answer:** millions
9. Coal is a **flammable** black hard rock used as a solid fuel. **Answer:** flammable
10. Coal is a **sedimentary** rock formed from peat. **Answer:** sedimentary
11. Coal is used in houses for **cooking** and heating. **Answer:** cooking
12. Coal is used to produce electricity in **power plants**. **Answer:** power plants
13. Coal is used in industries to produce steel, paper and **cement**. **Answer:** cement
14. Chemicals obtained from coal are used in manufacturing of dyes, **paints** and nylon. **Answer:** paints
15. Coal reserves in India are found in West Bengal, Odisha, Madhya Pradesh, Jharkhand, Andhra Pradesh, **Assam** and Tamil Nadu. **Answer:** Assam
16. Petroleum (crude oil) is a thick, **black** liquid fuel. **Answer:** black
17. Petroleum is obtained by drilling down holes through rocks on land or **off-shore**. **Answer:** off-shore
18. Petroleum has an **unpleasant** smell. **Answer:** unpleasant
19. Petrol (gasoline) and **diesel** are used for cars and other vehicles, and generators. **Answer:** diesel
20. **Jet fuel** is used for aeroplanes. **Answer:** Jet fuel
21. Kerosene and LPG are used as **domestic** fuels. **Answer:** domestic
22. Paraffin wax and lubricants like **grease** and motor oil are petroleum by-products. **Answer:** grease
23. **Bitumen** is used for construction of roads. **Answer:** Bitumen
24. Natural gas is a gaseous fossil fuel obtained along with petroleum in **oil-wells**. **Answer:** oil-wells
25. CNG (Compressed Natural Gas) is an **eco-friendly** fuel used in vehicles. **Answer:** eco-friendly

26. When we burn fuels, they release **carbon dioxide** gas that causes global warming. **Answer:** carbon dioxide
27. Burning of fossil fuels leads to melting of polar ice caps, flooding of low-lying areas and rise in sea **level**. **Answer:** level
28. Fossil fuels are **non-renewable** sources of energy. **Answer:** non-renewable
29. Alternative energy sources include solar energy, wind energy, hydroelectric energy and **tidal energy**. **Answer:** tidal energy
30. An **adventure** is an exciting or unusual experience accompanied with some bold and risky undertaking. **Answer:** adventure

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B. One-Word Answers (30 Marks)

1. What is any material that reacts with other substances to release heat and energy called? **Answer:** Fuel
2. Name one solid fuel. **Answer:** Coal / Wood / Coke
3. Name one liquid fuel. **Answer:** Petrol / Diesel / Kerosene
4. What is the gaseous fuel commonly used? **Answer:** Natural gas
5. What are fuels formed by natural decomposition of dead plants and animals called? **Answer:** Fossil fuels
6. Name the black hard rock used as a solid fuel. **Answer:** Coal
7. What type of rock is coal? **Answer:** Sedimentary
8. What is crude oil also called? **Answer:** Petroleum
9. What is the eco-friendly fuel used in vehicles? **Answer:** CNG
10. What gas is released when we burn fuels? **Answer:** Carbon dioxide
11. What is the full form of CNG? **Answer:** Compressed Natural Gas
12. What is the full form of LPG? **Answer:** Liquefied Petroleum Gas
13. Name one petroleum by-product used for road construction. **Answer:** Bitumen
14. Name one state in India where coal reserves are found. **Answer:** Jharkhand / Odisha / West Bengal
15. What is used to produce electricity in power plants? **Answer:** Coal
16. Name one alternative energy source. **Answer:** Solar energy / Wind energy
17. What causes global warming? **Answer:** Carbon dioxide
18. What type of energy source is fossil fuel? **Answer:** Non-renewable
19. What is drilling on the sea bed for petroleum called? **Answer:** Off-shore drilling
20. Name the fuel used for aeroplanes. **Answer:** Jet fuel
21. What is an exciting or unusual experience with risk called? **Answer:** Adventure

22. Name one adventure sport mentioned in the chapter. **Answer:** Mountaineering / Scuba diving
23. What is mountain climbing also known as? **Answer:** Mountaineering
24. What is climbing over rock, snow or ice called? **Answer:** Rock-craft
25. What is the highest peak in the world? **Answer:** Mt. Everest
26. Who was the first woman to climb Mt. Everest? **Answer:** Junko Tabei
27. Who was the first Indian woman to reach the peak of Mt. Everest? **Answer:** Bachendri Pal
28. Which woman has climbed Mt. Everest twice? **Answer:** Santosh Yadav
29. What is the area beyond Earth's atmosphere called? **Answer:** Space
30. Who was the first Indian to go into space? **Answer:** Rakesh Sharma

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C. True or False (30 Marks)

1. Fuels are used only for cooking. **Answer:** False (They are used for cooking, heating, running vehicles and generating electricity)
2. Coal is a liquid fuel. **Answer:** False (It is a solid fuel)
3. Petrol is a liquid fuel. **Answer:** True
4. Natural gas is a solid fuel. **Answer:** False (It is a gaseous fuel)
5. Fossil fuels are formed by natural processes. **Answer:** True
6. Fossil fuels take only a few years to form. **Answer:** False (They take millions of years to form)
7. Coal is a flammable rock. **Answer:** True
8. Coal is used to produce electricity. **Answer:** True
9. Petroleum is a clear liquid. **Answer:** False (It is thick and black)
10. Petroleum has a pleasant smell. **Answer:** False (It has an unpleasant smell)
11. Bitumen is used for construction of roads. **Answer:** True
12. Natural gas is found along with petroleum in oil-wells. **Answer:** True
13. CNG is harmful to the environment. **Answer:** False (It is an eco-friendly fuel)
14. Burning fuels releases oxygen. **Answer:** False (It releases carbon dioxide)
15. Fossil fuels cause air pollution. **Answer:** True
16. Fossil fuels are renewable sources of energy. **Answer:** False (They are non-renewable)
17. Solar energy is an alternative energy source. **Answer:** True
18. Wind energy is a fossil fuel. **Answer:** False (It is an alternative energy source)
19. Coal reserves in India are found in Jharkhand. **Answer:** True
20. Kerosene is used as a domestic fuel. **Answer:** True

21. Jet fuel is used for cars. **Answer:** False (It is used for aeroplanes)
22. Paraffin wax is a petroleum by-product. **Answer:** True
23. Mountaineering is not a risky sport. **Answer:** False (It is a thrilling and risky sport)
24. Mt. Everest is the highest peak in the world. **Answer:** True
25. Junko Tabei was the first woman to climb Mt. Everest. **Answer:** True
26. Santosh Yadav climbed Mt. Everest three times. **Answer:** False (She climbed it twice)
27. Space is the area beyond Earth's atmosphere. **Answer:** True
28. The Sun is a planet. **Answer:** False (It is a star)
29. Stars have their own light. **Answer:** True
30. Planets revolve around the Sun in fixed orbits. **Answer:** True

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D. Multiple Choice Questions (30 Marks)

1. Which of the following is a solid fuel? a) Petrol
   b) Coal
   c) Natural gas
   d) Kerosene
   Answer: b) Coal
2. Which of the following is a liquid fuel? a) Wood
   b) Coal
   c) Diesel
   d) Natural gas
   Answer: c) Diesel
3. What is an example of gaseous fuel? a) Petrol
   b) Coal
   c) Natural gas
   d) Wood
   Answer: c) Natural gas
4. Fossil fuels are formed from: a) Rocks
   b) Dead plants and animals
   c) Water
   d) Air
   Answer: b) Dead plants and animals
5. How long do fossil fuels take to form? a) Few days
   b) Few months

c) Few years

d) Millions of years

Answer: d) Millions of years

6. Coal is a: a) Liquid

b) Gas

c) Sedimentary rock

d) Metal

Answer: c) Sedimentary rock

7. What color is petroleum? a) White

b) Black

c) Blue

d) Red

Answer: b) Black

8. Which fuel is used for aeroplanes? a) Petrol

b) Diesel

c) Jet fuel

d) Kerosene

Answer: c) Jet fuel

9. What is used for construction of roads? a) Coal

b) Bitumen

c) Natural gas

d) Wood

Answer: b) Bitumen

10. CNG stands for: a) Compressed Natural Gas

b) Controlled Natural Gas

c) Carbon Natural Gas

d) Common Natural Gas

Answer: a) Compressed Natural Gas

11. Which state in India does NOT have coal reserves? a) Jharkhand

b) Odisha

c) Kerala

d) West Bengal

Answer: c) Kerala

12. What gas causes global warming? a) Oxygen

b) Nitrogen

c) Carbon dioxide

d) Hydrogen

Answer: c) Carbon dioxide

13. Fossil fuels are: a) Renewable

b) Non-renewable

c) Unlimited

d) Harmless

Answer: b) Non-renewable

14. Which is an alternative energy source? a) Coal

b) Petroleum

c) Solar energy

d) Natural gas

Answer: c) Solar energy

15. What is petroleum also called? a) Black gold

b) Crude oil

c) Liquid coal

d) Natural oil

Answer: b) Crude oil

16. LPG is used as: a) Vehicle fuel

b) Domestic fuel

c) Industrial fuel

d) Aeroplane fuel

Answer: b) Domestic fuel

17. Coal is used to produce: a) Water

b) Oxygen

c) Electricity

d) Food

Answer: c) Electricity

18. Which is NOT a fossil fuel? a) Coal

b) Petroleum

c) Natural gas

d) Solar energy

Answer: d) Solar energy

19. What is mountain climbing called? a) Hiking

b) Mountaineering

c) Trekking

d) Walking

Answer: b) Mountaineering

20. What is the highest peak in the world? a) K2
    b) Kanchenjunga
    c) Mt. Everest
    d) Nanda Devi
    Answer: c) Mt. Everest
21. How high is Mt. Everest? a) 7848m
    b) 8848m
    c) 9848m
    d) 6848m
    Answer: b) 8848m
22. Who was the first woman to climb Mt. Everest? a) Bachendri Pal
    b) Santosh Yadav
    c) Junko Tabei
    d) Kalpana Chawla
    Answer: c) Junko Tabei
23. Who was the first Indian woman to climb Mt. Everest? a) Junko Tabei
    b) Bachendri Pal
    c) Santosh Yadav
    d) Sunita Williams
    Answer: b) Bachendri Pal
24. Who climbed Mt. Everest twice? a) Bachendri Pal
    b) Junko Tabei
    c) Santosh Yadav
    d) Edmund Hillary
    Answer: c) Santosh Yadav
25. What is space? a) Area inside Earth
    b) Area beyond Earth's atmosphere
    c) Earth's surface
    d) Ocean floor
    Answer: b) Area beyond Earth's atmosphere
26. What is the Sun? a) Planet
    b) Moon
    c) Star
    d) Satellite
    Answer: c) Star
27. What do planets do around the Sun? a) Stand still
    b) Revolve in fixed orbits

c) Move randomly

d) Disappear

Answer: b) Revolve in fixed orbits

28. A group of stars is called: a) Galaxy

b) Constellation

c) Solar system

d) Universe

Answer: b) Constellation

29. What is a satellite? a) A star

b) A planet

c) A heavenly body that revolves around a planet

d) A fuel

Answer: c) A heavenly body that revolves around a planet

30. Who was the first Indian in space? a) Kalpana Chawla

b) Rakesh Sharma

c) Sunita Williams

d) Abdul Kalam

Answer: b) Rakesh Sharma

E. Define the Following Terms (20 Marks)

1. **Fuel Answer:** A fuel is any material that reacts with other substances to release heat and energy.
2. **Fossil Fuels Answer:** Fossil fuels are fuels that are formed by natural processes of decomposition of buried dead plants and animals. Coal, petroleum and natural gas are fossil fuels.
3. **Coal Answer:** Coal is a flammable black hard rock used as a solid fuel. It is a sedimentary rock formed from peat.
4. **Petroleum Answer:** Petroleum (crude oil) is a thick, black liquid fuel obtained by drilling holes through rocks on land or off-shore. It has an unpleasant smell.
5. **Natural Gas Answer:** Natural gas is a gaseous fossil fuel obtained along with petroleum in oil-wells. It is a domestic and industrial fuel.
6. **CNG Answer:** CNG (Compressed Natural Gas) is an eco-friendly fuel used in vehicles. It is a clean and cheap fuel.
7. **Sedimentary Rock Answer:** Sedimentary rock is a rock formed through the deposition and solidification of sediment. Coal is an example of sedimentary rock formed from peat.

8. **Bitumen Answer:** Bitumen is a viscous substance that exists in a liquid to a semi-solid phase. It is a petroleum by-product used for construction of roads.
9. **Non-renewable Energy Answer:** Non-renewable energy sources are those that cannot be replenished easily and take millions of years to form. Fossil fuels are non-renewable sources of energy.
10. **Alternative Energy Answer:** Alternative energy sources are renewable sources of energy that can be used instead of fossil fuels. Solar energy, wind energy, hydroelectric energy and tidal energy are examples.
11. **Adventure Answer:** An adventure is an exciting or unusual experience accompanied with some bold and risky undertaking. It includes activities like travelling, exploring, sky diving, mountaineering, scuba diving and river rafting.
12. **Mountaineering Answer:** Mountaineering is a sport of mountain climbing. Crossing glaciers or short stretches of rock climbing is known as hiking. It requires experience, athletic expertise and technical knowledge.
13. **Off-shore Drilling Answer:** Off-shore drilling is the process of extracting petroleum by drilling holes in the sea bed or ocean floor.
14. **Global Warming Answer:** Global warming is the increase in Earth's temperature caused by the release of carbon dioxide and other greenhouse gases when fossil fuels are burned.
15. **Flammable Answer:** Flammable means able to burn easily. Coal is a flammable rock.
16. **Constellation Answer:** A constellation is a group of stars in the space shuttle or vehicle in which humans travel into space and back again.
17. **Space Answer:** Space is the area beyond Earth's atmosphere. The heavenly bodies like Sun, moon, planets, meteors and asteroids are all parts of space.
18. **Satellite Answer:** A satellite is a heavenly body that revolves around a large body like a planet. Moon is Earth's natural satellite. Artificial satellites are machines created by man.
19. **Solar System Answer:** The Solar System consists of eight planets (Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus and Neptune) that revolve around the Sun in fixed orbits.
20. **Space Shuttle Answer:** A space shuttle is a vehicle in which humans travel into space and back again. It is used for space exploration.

F. Give Reasons (Answer in 2-3 lines) (15 Marks)

1. **Why are fossil fuels called non-renewable sources of energy? Answer:** Fossil fuels are called non-renewable sources of energy because they take millions of years to form and cannot be replenished easily. Once used, they cannot be replaced in a short time.

2. **Why should we not waste natural fuels? Answer:** We should not waste natural fuels because they are non-renewable and take millions of years to form. Once exhausted, they cannot be replaced, so we must use them carefully and wisely.
3. **Why should we use CNG whenever possible? Answer:** We should use CNG whenever possible because it is an eco-friendly fuel that produces less pollution. It is a clean and cheap fuel that helps reduce air pollution and protects the environment.
4. **Why do fossil fuels cause global warming? Answer:** Fossil fuels cause global warming because when we burn them, they release carbon dioxide gas. This gas traps heat in the atmosphere, leading to an increase in Earth's temperature and causing climate change.
5. **Why is coal called a fossil fuel? Answer:** Coal is called a fossil fuel because it is formed by natural processes of decomposition of buried dead plants and animals over millions of years. It is naturally occurring and takes a very long time to form.
6. **Why is petroleum refined? Answer:** Petroleum is refined because crude oil obtained from oil-wells contains many different substances mixed together. Refining separates these substances to obtain useful products like petrol, diesel, kerosene, and LPG.
7. **Why are alternative energy sources important? Answer:** Alternative energy sources are important because fossil fuels are non-renewable and will eventually run out. These renewable sources like solar and wind energy can provide unlimited energy without causing pollution or harming the environment.
8. **Why does burning of fossil fuels lead to rise in sea level? Answer:** Burning of fossil fuels leads to global warming which causes melting of polar ice caps and glaciers. When ice melts, the water flows into the oceans, causing the sea level to rise and flooding of low-lying coastal areas.
9. **Why is mountaineering considered a risky sport? Answer:** Mountaineering is considered risky because weather can change unexpectedly, equipment can fail, landslides can occur, and air pressure decreases at high altitudes causing breathlessness. Mountaineers can also fall due to inattention or other reasons.
10. **Why do astronauts carry oxygen cylinders in space? Answer:** Astronauts carry oxygen cylinders in space because there is no air or oxygen in space. Since humans need oxygen to breathe and survive, they must carry their own supply of oxygen when traveling in space.
11. **Why is the Sun important for life on Earth? Answer:** The Sun is important for life on Earth because it is the ultimate source of heat, light and energy. It provides warmth, helps plants make food through photosynthesis, and drives Earth's weather and climate systems.
12. **Why don't planets have their own light? Answer:** Planets don't have their own light because they are not stars. They are heavenly bodies that revolve around the Sun and only reflect the light they receive from the Sun.
13. **Why is coal used to produce electricity in power plants? Answer:** Coal is used to produce electricity in power plants because when it burns, it releases a large amount of heat energy. This heat is used to boil

water, create steam, and turn turbines that generate electricity.

14. **Why are satellites important? Answer:** Satellites are important because they help us forecast weather, send and receive wireless transmissions, locate minerals and resources in Earth's crust, and provide communication services. They also help in scientific research and navigation.
15. **Why must we feel breathless as we go to a higher altitude? Answer:** We feel breathless at higher altitudes because air pressure decreases with increasing height. There is less oxygen available in the air at high altitudes, making it difficult to breathe normally and causing breathlessness.

G. Short Questions / Write Short Notes (Answer in 4-5 lines) (10 Marks)

1. **What is a fuel? What are its different states? Answer:** A fuel is any material that reacts with other substances to release heat and energy. Fuels are used for cooking, heating, running vehicles and generating electricity. They exist in three states - solid fuels like wood, coal and coke; liquid fuels like petrol, diesel and kerosene; and gaseous fuels like natural gas. Fuels can be either natural or artificial.
2. **What are fossil fuels? Give examples. Answer:** Fossil fuels are fuels that are formed by natural processes of decomposition of buried dead plants and animals. Coal, petroleum and natural gas are naturally occurring fossil fuels. They take millions of years to form as dead organisms get buried under soil and rocks. Under high pressure and temperature, they gradually turn into fossil fuels.
3. **Write about coal and its uses. Answer:** Coal is a flammable black hard rock used as a solid fuel. It is a sedimentary rock formed from peat. Coal is used in houses for cooking and heating. It is used to produce electricity in power plants. It is used in industries to produce steel, paper and cement. Chemicals obtained from coal are used in manufacturing dyes, paints and nylon.
4. **How is petroleum formed? What are its by-products and their uses? Answer:** Petroleum is formed when dead sea animals get buried under Earth's surface. Under high temperature and pressure, they gradually turn into petroleum. Crude oil obtained from oil-wells is refined and the by-products obtained are: Petrol (gasoline) and diesel for vehicles and generators; Jet fuel for aeroplanes; Kerosene and LPG for domestic use; Paraffin wax and lubricants; and Bitumen for road construction.
5. **Give some uses of coal. Answer:** Coal has many important uses. It is used in houses for cooking and heating. It is used to produce electricity in power plants. It is used in industries to produce steel, paper and cement. Chemicals obtained from coal are used in the manufacturing of dyes, paints and nylon. Coal reserves in India are found in several states including West Bengal, Odisha, Madhya Pradesh, Jharkhand, Andhra Pradesh, Assam and Tamil Nadu.
6. **List some disadvantages of using fossil fuels. Answer:** When we burn fuels, they release carbon dioxide gas that causes global warming. This leads to melting of polar ice caps, flooding of low-lying areas and rise in sea level. Fossil fuels take millions of years to form and cannot be replenished easily,

making them non-renewable sources of energy. Gases released from burning fossil fuels cause air pollution. We need to look for alternative energy sources to reduce our dependence on fossil fuels.

7. **What are alternative energy sources? Give examples. Answer:** Since fossil fuels are non-renewable sources of energy, we need to look for alternative energy sources. Alternative sources include solar energy, wind energy, hydroelectric energy and tidal energy. Solar cookers and windmills are examples of devices that use alternative energy. These sources are renewable, meaning they can be used repeatedly without running out. They are also environment-friendly as they don't cause pollution.
8. **What is mountaineering? Write about its challenges. Answer:** Mountaineering is a sport of mountain climbing. Crossing glaciers or short stretches of rock climbing is known as hiking. Mountaineering done over rock, snow or ice is called rock-craft, snow-craft or skiing respectively. Mountain climbing is thrilling as well as risky. Weather can change unexpectedly, equipment can fail, and landslides can occur. Air pressure decreases with increasing altitude causing breathlessness and tiredness. Mountaineers always carry oxygen cylinders with them.
9. **Write a short note on Mt. Everest. Answer:** Mt. Everest is the highest peak (8848m) in the world and has been the centre of attraction for all climbers. Sir Edmund Hillary and Sherpa Tenzing Norgay were the first to scale Mt. Everest. Junko Tabei was the first woman to climb it. Bachendri Pal was the first Indian woman to reach the peak of Mt. Everest. Santosh Yadav is the only woman in the world to climb Mt. Everest twice.
10. **What is a satellite? What are the different types of satellites? Answer:** A satellite is a heavenly body that revolves around a large body like a planet. Moon is Earth's natural satellite. Satellites can be natural or artificial. Artificial satellites are machines created by man that are set up in space to forecast weather, send and receive wireless transmissions, and locate minerals and other resources. Planets like Mercury have no moon whereas Jupiter has 67 moons.

End of Worksheet